



VOLTAIRE

ENERGY MARKETS INTELLIGENCE

Energy Markets Intelligence **Terminal**

A real-time crude & refined-products analytics platform — covering market structure, refining economics, fundamentals, positioning, weather-driven demand and sentiment for the global oil complex.

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PROJECT LINKS

Live App huggingface.co/spaces/HarshJ12/Oil-Market-Dashboard

GitHub github.com/HarshJamgaonkar/Energy-Market-Dashboard

Project Explanation & Submission
Financial & Analytical Methodology

Instruments: Brent · WTI · Heating Oil · RBOB · Gas Oil
Live · Dataset · Official feeds

Contents

1	Executive Summary	Overview & objectives
2	The Energy Complex	Instruments, benchmarks & units
3	Data Sources	Every feed, cadence & coverage
4	Market Structure	Forward curves, term structure, calendar spreads
5	Refining Economics	Crack spreads & margins
6	Inter-Commodity Spreads	Arbitrage relationships
7	Correlation Structure	Rolling return correlation
8	Seasonality	Calendar-month behaviour
9	Statistical Context	Percentiles & z-scores
10	Fundamentals & Inventories	Stocks, the 5-yr band, refining, rigs
11	Speculative Positioning	CFTC Commitments of Traders
12	Weather & Demand	Degree days → product demand
13	Macro Backdrop	Dollar, equities, vol, rates
14	News & Sentiment	FinBERT financial NLP
15	Alerts Engine	Thresholds & anomaly watch
16	Unit Conventions	Conversions & reference
17	Data Provenance	Methodology & integrity policy
18	Technology Stack	Architecture (brief)
19	Glossary	Key terms

1 Executive Summary

Voltaire Terminal is an institutional-style intelligence dashboard for the global oil complex. It consolidates the four questions an energy trader asks every morning — *where are prices, what is the market structure telling me, what do the fundamentals say, and how is everyone positioned* — into a single, continuously-updating screen.

The platform tracks five core instruments spanning the crude-to-products value chain: two crude benchmarks (Brent and WTI) and three refined products (Heating Oil/ULSD, RBOB Gasoline and ICE Gas Oil). Around these it computes the derived relationships that actually drive trading decisions: **forward-curve structure** (contango/backwardation), **crack spreads** (refining margins), **inter-commodity arbitrages**, **cross-sectional correlation**, **seasonality**, and **statistical richness/cheapness**.

It layers physical fundamentals on top — EIA petroleum inventories with a five-year seasonal envelope, refinery utilisation, rig counts and weekly builds/draws — and contextual drivers: speculative positioning (CFTC Commitments of Traders), weather-driven demand (heating/cooling degree days), the cross-asset macro backdrop, and a financial-language model (FinBERT) scoring a live newswire.

DESIGN PRINCIPLE — PROVENANCE OVER GUESSWORK

Every number on the screen is traceable to a named source and tagged by kind — **LIVE** **DATASET** **OFFICIAL** **DERIVED** **MODELLED**. Where no free, open source exists for a value, the platform shows an em-dash (—) rather than a fabricated figure. This integrity policy is central to the project.

What makes the analytics genuine

Live prices reflect the real market in real time (front-month futures), while multi-year history is sourced from a proprietary **1-minute forward-curve dataset** spanning 2021–2026. This separation lets the dashboard be simultaneously *live* (today's prices) and *deep* (years of real term-structure history for backtesting structure, cracks, correlation and seasonality).

2 The Energy Complex Covered

The oil market is not one price but a *chain* of related prices: crude oil is extracted, transported, and refined into products (gasoline, distillates) that are consumed. The economics of every link — the spread between crude and product, between regions, between delivery months — is where information and opportunity live. Voltaire covers both ends of the chain.

INSTRUMENT	SYMBOL	VENUE	QUOTE UNIT	ROLE IN THE COMPLEX
Brent Crude	BZ=F / LCO	ICE	\$/barrel	The global crude benchmark; prices ~two-thirds of internationally traded crude.
WTI Crude	CL=F / CL	NYMEX	\$/barrel	The U.S. crude benchmark, priced at Cushing, Oklahoma.
Heating Oil (ULSD)	HO=F / HO	NYMEX	\$/gallon	Middle distillate; proxy for diesel & jet. Winter-demand sensitive.
RBOB Gasoline	RB=F	NYMEX	\$/gallon	Light product; the U.S. gasoline benchmark. Summer-driving-season sensitive.
ICE Gas Oil	LGO	ICE	\$/tonne	The European distillate benchmark (diesel/heating). Priced per metric tonne.

Why these five

Together they form a complete, tradable refining picture: two crudes that set the global and U.S. price of feedstock, two U.S. products (gasoline + distillate) that define the American refining margin, and the European distillate benchmark that completes the transatlantic distillate arbitrage. Their interplay produces every spread the terminal computes.

A NOTE ON UNITS

Crudes trade in **dollars per barrel**, U.S. products in **dollars per gallon**, and Gas Oil in **dollars per metric tonne**. To compare or combine them (e.g. to compute a refining margin) everything must first be converted to a common \$/barrel basis. Those conversions — ×42 for gallons, ÷7.45 for Gas Oil tonnes — are detailed in §16 and applied throughout.

3 Data Sources

The platform is wired to eight independent sources. Each was chosen to be **free and open** (no paywall) while remaining authoritative. The table summarises coverage; detailed notes follow.

SOURCE	KIND	CADENCE	FEEDS
Yahoo Finance Futures chart API	LIVE	~Real-time (45s cache)	Front-month quotes & daily history for all five instruments + macro; the live anchor for forward curves; RBOB's real dated-contract curve.
Proprietary 1-min forward-curve dataset	DATASET	Historical 2021–2026	Multi-year daily price history, the real forward-curve term structure (M1–M12), correlation, seasonality and crack-spread history.
EIA U.S. Energy Information Administration, Open Data v2	OFFICIAL	Weekly / Monthly	Crude, Cushing, SPR, gasoline & distillate stocks; PADD breakdown; refinery utilisation; official spot (RWTC/RBRTE); rig count; weekly builds/draws.
Open-Meteo	LIVE	Hourly	Regional temperatures & the heating/cooling degree-day demand model.
NOAA NHC + CPC	OFFICIAL	30 min / 12 h	Active tropical-cyclone tracker (Gulf supply risk) & the El Niño/La Niña (ENSO/ONI) index.
CFTC Commitments of Traders	OFFICIAL	Weekly	Managed-money (speculative) net positioning for WTI, Brent, RBOB and ULSD.
Financial Juice	LIVE	Real-time RSS	The breaking-news wire feeding the news panel & sentiment.
FinBERT ProsusAI/finbert	DERIVED	On each headline	A financial-language model scoring every headline positive / negative / neutral.

3.1 Yahoo Finance — the live tape LIVE

A single public chart endpoint returns both the live quote and the historical close series for any futures symbol. It powers the price tiles, the ticker, the movers ranking, the macro block, and — crucially — provides the **live front-month price** that anchors every forward curve. For RBOB (absent from the proprietary dataset) it also supplies a genuine forward curve assembled from individual dated contracts (e.g. RBN26, RBQ26, ...).

3.2 The proprietary forward-curve dataset DATASET

The analytical backbone of the project. Five large CSV files capture the **full forward-curve term structure at one-minute resolution** for WTI, Brent, Heating Oil and Gas Oil from January 2021 to May 2026. Each row is a timestamp carrying, for each of up to fourteen consecutive contract months (c1...c14), the contract code and its volume-*weighted mid* price.

Because raw size is multiple gigabytes, a streaming pipeline distils each file *once* into a compact artefact containing (a) the **daily front-month close** (the last c1 mid of each trading day) and (b) the **latest full**

forward curve (the most recent M1–M12 settlements). This gives years of real history — and real term structure — at a fraction of the footprint.

WHY THIS MATTERS FINANCIALLY

Most free data gives you only a continuous front-month price. This dataset gives the *whole curve over time* — so the terminal can show the genuine shape of contango/backwardation, real calendar spreads, and a true multi-year history of refining margins, rather than modelled approximations.

3.3 EIA — the fundamentals of supply OFFICIAL

The U.S. Energy Information Administration publishes the most closely-watched supply data in the oil market. The Weekly Petroleum Status Report (every Wednesday) gives commercial crude, Cushing, Strategic Petroleum Reserve, gasoline and distillate stocks, plus refinery utilisation. The platform pulls these for the inventory panels, builds the five-year seasonal band (§10), and reads the official Cushing/Brent spot series and the monthly rig count.

3.4 Open-Meteo — weather & demand LIVE

Hourly-refreshed temperature forecasts for six representative demand regions (U.S. Northeast/Midwest/Gulf, NW & S Europe, NE Asia). From these the terminal computes heating- and cooling-degree-days and a product-demand signal (§12).

3.5 NOAA — physical risk OFFICIAL

The National Hurricane Center's active-storm feed flags tropical cyclones that threaten Gulf of Mexico production and refining (a recurring supply shock), and the Climate Prediction Center's Oceanic Niño Index frames the seasonal weather regime (El Niño / La Niña), which conditions heating and cooling demand months ahead.

3.6 CFTC — who is positioned how OFFICIAL

The weekly Commitments of Traders report decomposes open interest by trader type. The terminal isolates **Managed Money** (hedge funds / speculators) net positioning — the cohort whose crowding and capitulation drive short-term price swings (§11).

3.7 Financial Juice + FinBERT — the narrative LIVE DERIVED

A real-time financial newswire supplies headlines; **FinBERT** — a BERT language model fine-tuned on financial text — classifies each one positive / negative / neutral. This converts an unstructured news stream into a quantified sentiment tone (§14).

4 Market Structure — Forward Curves & Term Structure

A futures forward curve plots the price of the same commodity for delivery in successive future months (M1, M2, ... M12). Its *shape* is one of the most information-rich objects in commodities: it encodes the market's read on the balance between supply, demand and storage.

Contango vs. backwardation

- **Backwardation** — front months priced *above* deferred months (a downward-sloping curve). Signals a **tight** physical market: buyers pay a premium for prompt barrels. Bullish; rewards holding length (positive roll yield).
- **Contango** — front months priced *below* deferred months (an upward-sloping curve). Signals **surplus**: prompt barrels are abundant and the market pays to store them. Bearish; penalises long positions via negative roll.

How the curve is constructed

The terminal keeps the front month **live** while sourcing the realistic **shape** from the dataset's observed term structure. The dataset curve is parallel-shifted so that M1 sits exactly on today's live front price, preserving the real month-to-month spreads:

FORWARD CURVE CONSTRUCTION

$$V(M_i) = F_{\text{live}} + (C(M_i) - C(M_1))$$

where F_{live} = live front-month price (Yahoo)
 $C(M_i)$ = dataset settlement of contract month i
 $C(M_1)$ = dataset front-month settlement

This yields a curve that reflects *today's price level* with a *genuinely observed structure*. RBOB is the exception: its curve is assembled directly from live Yahoo dated contracts (M1–M12 = consecutive real settlements), so it is fully live.

Calendar (time) spreads

A calendar spread is the price difference between two delivery months of the same commodity — the purest expression of curve shape, isolating structure from outright price direction.

CALENDAR SPREADS

$$\begin{aligned} M1-M2 &= V(M_1) - V(M_2) && \text{(prompt structure)} \\ M2-M3 &= V(M_2) - V(M_3) \\ M1-M6 &= V(M_1) - V(M_6) && \text{(half-year structure)} \\ M1-M12 &= V(M_1) - V(M_{12}) && \text{(annual structure)} \end{aligned}$$

Positive $\Rightarrow$ backwardation $\Rightarrow$ tightness (bullish)

Negative $\Rightarrow$ contango $\Rightarrow$ surplus (bearish)

TRADER'S READ The prompt spread (M1–M2) is the market's most sensitive tightness gauge; the annual spread (M1–M12) reflects structural balance and storage economics. A curve flipping from contango to backwardation

is a classic bullish regime signal.

5 Refining Economics — Crack Spreads

A **crack spread** is the gross margin a refiner earns by "cracking" a barrel of crude into refined products. It is the single most important profitability measure in refining and a key demand signal: when product demand is strong relative to crude, cracks widen and refiners run harder, pulling on crude.

The common \$/barrel basis

Because crude is priced per barrel but products per gallon (or tonne), every leg is first converted to \$/barrel before differencing:

UNIT NORMALISATION TO \$/BBL

Heating Oil, RBOB : price(\$/gal) × 42 (42 US gallons per barrel)

Gas Oil : price(\$/mt) ÷ 7.45 (≈7.45 barrels per tonne)

Crude (WTI/Brent) : already \$/bbl

Single-product cracks

PRODUCT CRACK (\$/BBL)

Crack = Product(\$/bbl) - Crude(\$/bbl)

e.g. Heating-Oil crack vs WTI = HO×42 - WTI

Gas-Oil crack vs Brent = GasOil÷7.45 - Brent

Refinery-yield cracks (the headline margins)

Real refineries produce a *slate* of products from each barrel, dominated by gasoline and distillate. Industry-standard blended cracks weight the products in proportions that approximate a typical refinery yield. The flagship is the **3:2:1**: three barrels of crude yield roughly two of gasoline and one of distillate.

BLENDED REFINING MARGINS (\$/BBL)

3:2:1 crack = (2·RBOB + 1·HO - 3·Crude) / 3 ← headline U.S. margin

2:1:1 crack = (1·RBOB + 1·HO - 2·Crude) / 2

5:3:2 crack = (3·RBOB + 2·HO - 5·Crude) / 5

(all product legs first converted to \$/bbl via ×42)

CRACK	COMPOSITION	WHAT IT SIGNALS
3:2:1	2 gasoline + 1 distillate vs 3 crude	The benchmark U.S. refining margin; refiner profitability & product demand.
RBOB crack	Gasoline – crude	Gasoline-specific margin; peaks into summer driving season.
HO / Gas-Oil crack	Distillate – crude	Diesel/heating margin; firms into winter & on industrial demand.

Real margin history & statistical context

The terminal evaluates each crack formula *day-by-day across years* of dataset prices (RBOB-based cracks use live Yahoo daily history), producing a true ~1-year margin history rather than a synthetic placeholder. Each live crack is then placed in its own historical distribution via a percentile and z-score (§9) — answering "is this margin rich or cheap versus its own past?".

TRADER'S READ Widening cracks ⇒ strong product demand, bullish refiners, supportive of crude runs.

Collapsing cracks ⇒ demand destruction or product glut; refiners cut runs, easing crude demand.

6 Inter-Commodity Spreads

Beyond the crude→product crack, the relationships *between* instruments encode geography, quality and seasonal substitution. These arbitrage spreads are where relative-value trades live.

SPREAD	DEFINITION	ECONOMIC MEANING
Brent – WTI	Brent – WTI (\$/bbl)	The transatlantic crude arbitrage. Driven by U.S. vs. international supply/demand, pipeline & shipping economics and quality. A wide Brent premium incentivises U.S. crude exports.
RBOB – Heating Oil	RBOB – HO (\$/gal)	The "gas–heat" spread; the seasonal rotation between gasoline (summer) and distillate (winter) demand.
Heating Oil – Gas Oil	HO×42 – GasOil÷7.45	The transatlantic distillate arbitrage (U.S. ULSD vs European Gas Oil); drives cross-Atlantic diesel cargo flows.
RBOB – Gas Oil	RBOB×42 – GasOil÷7.45	Light-vs-middle product value across regions.

All spreads are computed live from the price tape; where legs are in different units they are normalised to a common basis first (\$16).

NOTE — GAS OIL CALIBRATION Gas Oil has no free live feed, so its live price is derived from NYMEX ULSD and then *calibrated to the dataset's observed ULSD↔Gas-Oil ratio*, restoring a realistic distillate spread (a naïve unit conversion would otherwise collapse the Heating-Oil – Gas-Oil spread to zero).

7 Correlation Structure

Correlation quantifies how tightly two instruments move together — essential for hedging (which product best hedges which crude?), diversification, and reading regime. The terminal computes a 30-day rolling correlation matrix from **daily returns** (not prices, which trend and would inflate correlation).

DAILY SIMPLE RETURN

$$r(t) = (P(t) - P(t-1)) / P(t-1)$$

PEARSON CORRELATION OVER THE TRAILING 30 DAYS

$$\rho(X,Y) = \frac{\sum (r_x - \bar{r}_x) (r_y - \bar{r}_y)}{\sqrt{ \sum (r_x - \bar{r}_x)^2 } \cdot \sqrt{ \sum (r_y - \bar{r}_y)^2 } }$$

$\rho \in [-1, +1]$: $+1$ = move in lockstep, 0 = unrelated, -1 = opposite

Interpretation

Crude benchmarks (Brent–WTI) typically correlate near $+0.9$; the two middle distillates (Heating Oil and Gas Oil) are the tightest product pair. Crude–distillate correlation *loosens* whenever refining margins move on their own — a signal that product-specific (rather than crude-driven) forces are at work, which is itself tradable information.

8 Seasonality

Energy is the most seasonal of all commodity complexes: gasoline demand peaks in the summer driving season, heating fuels into winter. Seasonality answers two questions — what does each instrument *typically* do each calendar month, and where are we versus that typical path?

Price seasonality

For each calendar month, the terminal averages the month-over-month return across all available years, and records how often that month was positive (the "hit rate" — a reliability gauge).

PER CALENDAR MONTH M

$$\text{AvgReturn}(m) = \text{mean}_{\text{all years}} [P(m) / P(m-1) - 1]$$
$$\text{HitRate}(m) = \% \text{ of years in which that month's return was positive}$$

Cumulative "typical path", indexed to 100 at year-start:

$$\text{Path}(m) = \text{Path}(m-1) \times (1 + \text{AvgReturn}(m))$$

Crack seasonality

Refining margins are even more seasonal than outright prices. For the distillate cracks, the terminal averages the crack *level* (\$/bbl) for each calendar month and compares the latest value to its monthly norm — quantifying whether the current margin is seasonally rich or cheap.

TRADER'S READ Seasonality is a probabilistic backdrop, not a guarantee — a high hit rate (e.g. "this month was up in 8 of 10 years") raises the prior, which combines with structure, fundamentals and positioning to form a view.

9 Statistical Context — Percentiles & Z-Scores

A raw number ("the 3:2:1 crack is \$46") means little without context. Is \$46 high or low? The terminal answers by placing every live spread/crack within its own historical distribution — turning a number into a *signal*.

PERCENTILE RANK OF THE CURRENT VALUE WITHIN ITS HISTORY

$$\text{Percentile} = (\text{count of historical values} \leq \text{current}) / N \times 100$$

e.g. 85th percentile $\Rightarrow$ richer than 85% of the past year

Z-SCORE (STANDARD DEVIATIONS FROM THE MEAN)

$$z = (\text{current} - \mu) / \sigma$$

where μ = mean and σ = standard deviation of the history

$$|z| \geq 1.5\sigma \quad \text{or} \quad \text{percentile} \geq 85 / \leq 15 \quad \Rightarrow \quad \text{statistically stretched}$$

These two complementary lenses — percentile (distribution-free, robust to outliers) and z-score (parametric, scaled by volatility) — flag mean-reversion candidates and regime extremes. A crack at the 90th percentile and $+1.8\sigma$ is historically expensive; the burden of proof shifts to "why should it stay there?".

10 Fundamentals & Inventories

Prices are set at the margin by the physical balance of supply and demand, and the clearest read on that balance is **inventories**. The terminal surfaces the official EIA stock data with the context that makes it actionable.

Stocks vs. the five-year seasonal band

An absolute inventory number is meaningless without a seasonal reference — stocks are *supposed* to build in some months and draw in others. The terminal builds an envelope from the **five prior calendar years**: for each week-of-year it takes the minimum, maximum and average across those years (the current year excluded, so the comparison is honest).

SEASONAL BAND, PER WEEK-OF-YEAR W

Band(w) = [min , max] across the 5 prior years' week-w readings

Avg5y(w) = mean across the 5 prior years' week-w readings

Current below the band $\Rightarrow$ tight (bullish)

Current above the band $\Rightarrow$ surplus (bearish)

Weekly builds & draws

The week-on-week change in crude stocks is the market's most-watched number. A **draw** (inventories fall) signals demand outpacing supply — bullish; a **build** is bearish. The terminal surfaces the real EIA crude, gasoline and distillate changes and feeds them into the economic-release calendar's "prior" column.

Refinery utilisation & rig count

- **Refinery utilisation** — the % of operable capacity running. High utilisation means strong product output (and strong crude demand); a sharp drop (turnarounds, outages) tightens products and loosens crude.
- **Rig count** — the EIA monthly count of rotary rigs drilling for oil & gas. Drilling today becomes production months later, so the rig trend is a *leading* supply indicator. (Baker Hughes publishes the weekly figure but with no free API; the platform uses EIA's official monthly series instead.)

11 Speculative Positioning — CFTC Commitments of Traders

Prices are moved in the short run not only by fundamentals but by *flows* — and the largest discretionary flow comes from Managed Money (hedge funds, CTAs). The weekly CFTC report reveals how this cohort is positioned, which is both a momentum signal and a contrarian one at extremes.

MANAGED-MONEY POSITIONING METRICS

Net	= Long - Short	(net length in contracts)
Wow change	= Net(this week) - Net(last week)	(positioning momentum)
3-yr %ile	= rank of current Net within ~3 years of history	
Long share	= Long / (Long + Short)	

CONTRARIAN EXTREMES A net long in the $\geq$ **85th percentile** is "crowded long" — positioning is stretched, vulnerable to a long-liquidation flush (contrarian bearish). A net in the $\leq$ **15th percentile** is "crowded short" — primed for a short-covering squeeze (contrarian bullish). The week-on-week change shows whether money is currently flowing in or out.

The terminal tracks this for WTI, Brent, RBOB and ULSD, with the net-position history charted and percentile/extreme flags computed automatically.

12 Weather & Demand

Weather is a primary, near-term demand driver for energy: cold lifts heating-fuel (distillate, natural gas) demand; heat lifts cooling/power demand. The standard quantification is the **degree day**, measuring how far temperature deviates from a comfort baseline ($\approx 18^\circ\text{C}$ / 65°F).

DEGREE DAYS (PER DAY, PER REGION)

```
HDD = max( 0 , 18°C - T_mean )      heating demand
CDD = max( 0 , T_mean - 18°C )      cooling demand
(T_mean = daily mean temperature)
```

From weather to a product-demand signal

The terminal aggregates actual versus *climatologically-normal* degree days across all regions, expresses the gap as an anomaly, and translates it into a demand read:

DEGREE-DAY DEMAND ANOMALY

$$\text{Anomaly\%} = (\Sigma \text{ actual DD} - \Sigma \text{ normal DD}) / \Sigma \text{ normal DD} \times 100$$

Heating (HDD-driven) $\rightarrow$ Heating Oil $\cdot$ Nat Gas

Cooling (CDD-driven) $\rightarrow$ Power burn $\cdot$ Gasoline

colder than normal $\Rightarrow$ bullish heating fuels

hotter than normal $\Rightarrow$ bullish cooling/power demand

WHY THE SIGNAL IS GATED A large *percentage* anomaly off a tiny off-season base is statistically meaningless (heating demand in July is negligible regardless of the percentage). The signal is therefore **gated by absolute load**: it reads neutral until the absolute degree-day total is material, preventing misleading off-season readings.

13 Macro Backdrop

Oil does not trade in isolation; it sits inside a cross-asset macro context that conditions risk appetite and the dollar value of a barrel. The terminal tracks four reference assets.

INDICATOR	SYMBOL	LINK TO OIL
U.S. Dollar Index	DXY	Oil is priced in dollars, so a stronger dollar tends to weigh on oil (and vice-versa) — typically an inverse relationship.
S&P 500	SPX	Broad risk appetite / global growth proxy; risk-on phases generally support cyclical commodities like oil.
Volatility Index	VIX	The market's "fear gauge"; spikes accompany risk-off de-leveraging that often spills into commodities.
10-Year Treasury Yield	UST10Y	The rates/growth backdrop; influences the discount rate, the dollar and demand expectations.

14 News & Sentiment — Financial NLP

Markets react to narrative. To quantify it, the terminal scores a live newswire with **FinBERT** — a BERT language model fine-tuned on financial text — which classifies each headline into positive / negative / neutral with a probability for each class.

PER-HEADLINE SIGNED SCORE

$$\text{signed} = P(\text{positive}) - P(\text{negative}) \quad \in [-1 , +1]$$

AGGREGATE NEWS-TONE INDEX (0–100, 50 = NEUTRAL)

$$\text{Index} = 50 + 50 \times \text{mean}(\text{signed}) \quad \text{over the scored headlines}$$

> 55 bullish tone 45–55 neutral < 45 bearish tone

A second, price-based sentiment gauge maps each instrument's daily percentage move onto the same 0–100 scale, giving a momentum read per sub-sector (crude / distillates / gasoline / gas oil). Together, narrative tone and price momentum form a compact sentiment picture. If the language model is ever unavailable, the system transparently falls back to a keyword heuristic rather than failing.

15 Alerts Engine

The terminal watches the market so the user doesn't have to. Two complementary mechanisms run continuously over live metrics (prices, daily moves, cracks, curve slopes):

- **Threshold rules** — user-defined "alert me when X crosses Y". These are *edge-triggered*: they fire once on the crossing and re-arm only after the metric crosses back, avoiding repeated nuisance alerts.
- **Drastic-move auto-watch** — an always-on anomaly detector on the major instruments and the headline 3:2:1 crack, flagging outsized intraday moves against a rolling look-back, with per-metric cool-downs and an adjustable sensitivity.

Alerts surface as a notification badge, a toast and (optionally) a browser notification — turning the dashboard from a passive screen into an active monitor.

16 Unit Conventions & Conversions

Combining instruments quoted in different units is the source of most errors in energy analytics. The terminal standardises everything to **\$/barrel** for any cross-instrument calculation.

QUANTITY	CONVERSION	BASIS
Gallons → barrels	$\$/gal \times 42 = \$/bbl$	1 barrel = 42 US gallons (Heating Oil, RBOB)
Tonnes → barrels (Gas Oil)	$\$/mt \div 7.45 = \$/bbl$	≈ 7.45 barrels per metric tonne (density of gas oil)
Gas Oil $\$/gal \rightarrow \$/mt$ (proxy)	$\$/gal \times 42 \times 7.45 \times k$	ULSD proxy $\times$ calibration factor k (dataset-derived)
Crude	already $\$/bbl$	WTI, Brent quoted per barrel

The calibration factor $k = (\text{latest real Gas Oil}) \div (\text{latest ULSD converted by pure units})$, taken from the dataset, corrects the systematic gap between a naïve unit conversion and the real Gas Oil price so that derived distillate spreads stay realistic.

17 Data Provenance & Methodology

A core commitment of the project is that **every value is honestly sourced**. Each number carries a provenance kind, visible on hover, and the dashboard never presents a fabricated figure as if it were real.

KIND	MEANING	EXAMPLES
LIVE	Fetches fresh from a public feed	Futures quotes, weather, newswire
DATASET	From the proprietary forward-curve history	Price history, curve structure, crack history
OFFICIAL	Government / exchange data	EIA, NOAA, CFTC
DERIVED	Computed on the server from live feeds	Spreads, cracks, correlations, sentiment
MODELLED	No free source exists — shown as "—", not invented	Tanker freight, OPEC spare capacity, tank-level storage

The "em-dash" integrity policy

Where a metric has no free, open data source (for example, real-time tanker freight rates or satellite tank-level storage, which are paywalled), the platform displays an em-dash (—) and a clear "no source" label rather than a plausible-looking but invented number. This makes the boundary between real and unavailable data explicit — a discipline borrowed from institutional research standards.

Live vs. historical separation

Current prices always come from the live tape (Yahoo); multi-year history always comes from the dataset. The forward curve bridges the two — a live front month carried along a real, observed term structure — so the screen is simultaneously current and deep.

18 Technology Stack (brief)

Although this document focuses on the financial methodology, the implementation in outline:

LAYER	TECHNOLOGY
Frontend	React + Vite, Tailwind CSS, Recharts (charting), Framer Motion
Backend / API	Node.js + Express — aggregates every source behind a single /api and tags each response with its provenance
Machine learning	FinBERT via Transformers.js (ONNX) — runs in-process, no external inference service
Data pipeline	A streaming Node script distils the multi-gigabyte 1-minute CSVs into a compact committed artefact (daily closes + latest curves)
Resilience	Aggressive caching, last-good fallbacks, and graceful per-panel degradation — a flaky upstream never blanks the screen
Deployment	Single-image Docker, hosted on Hugging Face Spaces (UI + API + model on one origin)

19 Glossary

Backwardation	Forward curve sloping down — prompt prices above deferred; signals tightness.
Contango	Forward curve sloping up — prompt below deferred; signals surplus / storage incentive.
Crack spread	Refining margin: the value of products minus the crude used to make them, in \$/bbl.
3:2:1	The benchmark blended crack: 3 crude → 2 gasoline + 1 distillate.
Calendar spread	Price difference between two delivery months of the same commodity.
Roll yield	The return earned (or paid) as a futures position "rolls" down (or up) the curve over time.
COT	CFTC Commitments of Traders — weekly report of positioning by trader type.
Managed Money	Hedge funds / CTAs — the speculative cohort whose flows drive short-term moves.
HDD / CDD	Heating / Cooling Degree Days — temperature deviation from an ~18 °C baseline; a demand proxy.
PADD	Petroleum Administration for Defense District — the five U.S. regions EIA reports stocks by.
SPR	Strategic Petroleum Reserve — U.S. government emergency crude stockpile.
ENSO / ONI	El Niño–Southern Oscillation / Oceanic Niño Index — the seasonal climate regime.
ULSD	Ultra-Low-Sulphur Diesel — the modern Heating Oil contract; a middle distillate.
RBOB	Reformulated Blendstock for Oxygenate Blending — the U.S. gasoline futures benchmark.
FinBERT	A BERT language model fine-tuned on financial text for sentiment classification.